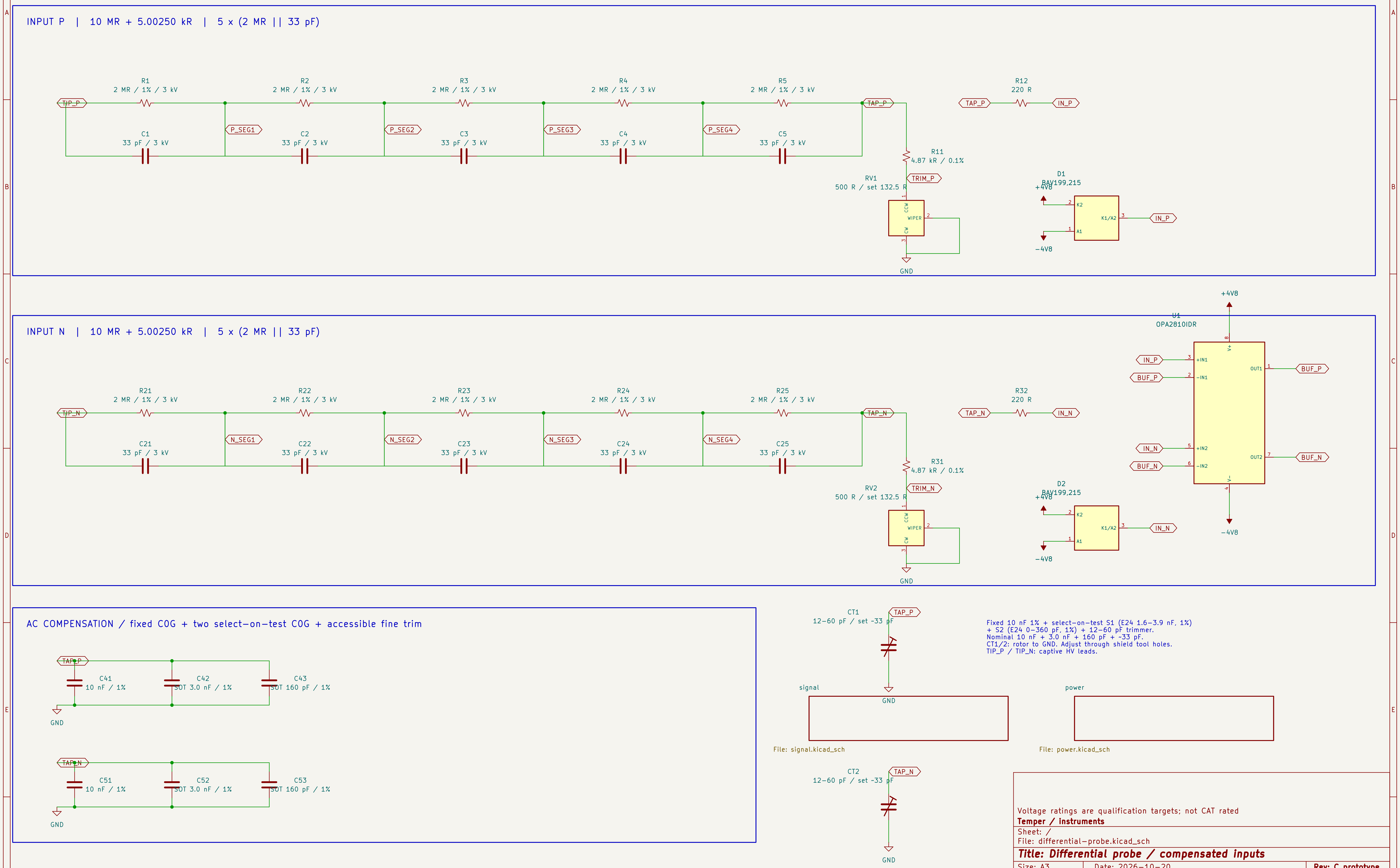
2000:1 INPUTS / mirror–matched geometry about the inter–leg slot, fixed grounded shield



INPUT P | 10 MR + 5.00250 kR | 5 x (2 MR || 33 pF)

INPUT N | 10 MR + 5.00250 kR | 5 x (2 MR || 33 pF)

AC COMPENSATION / fixed C0G + two select–on–test C0G + accessible fine trim

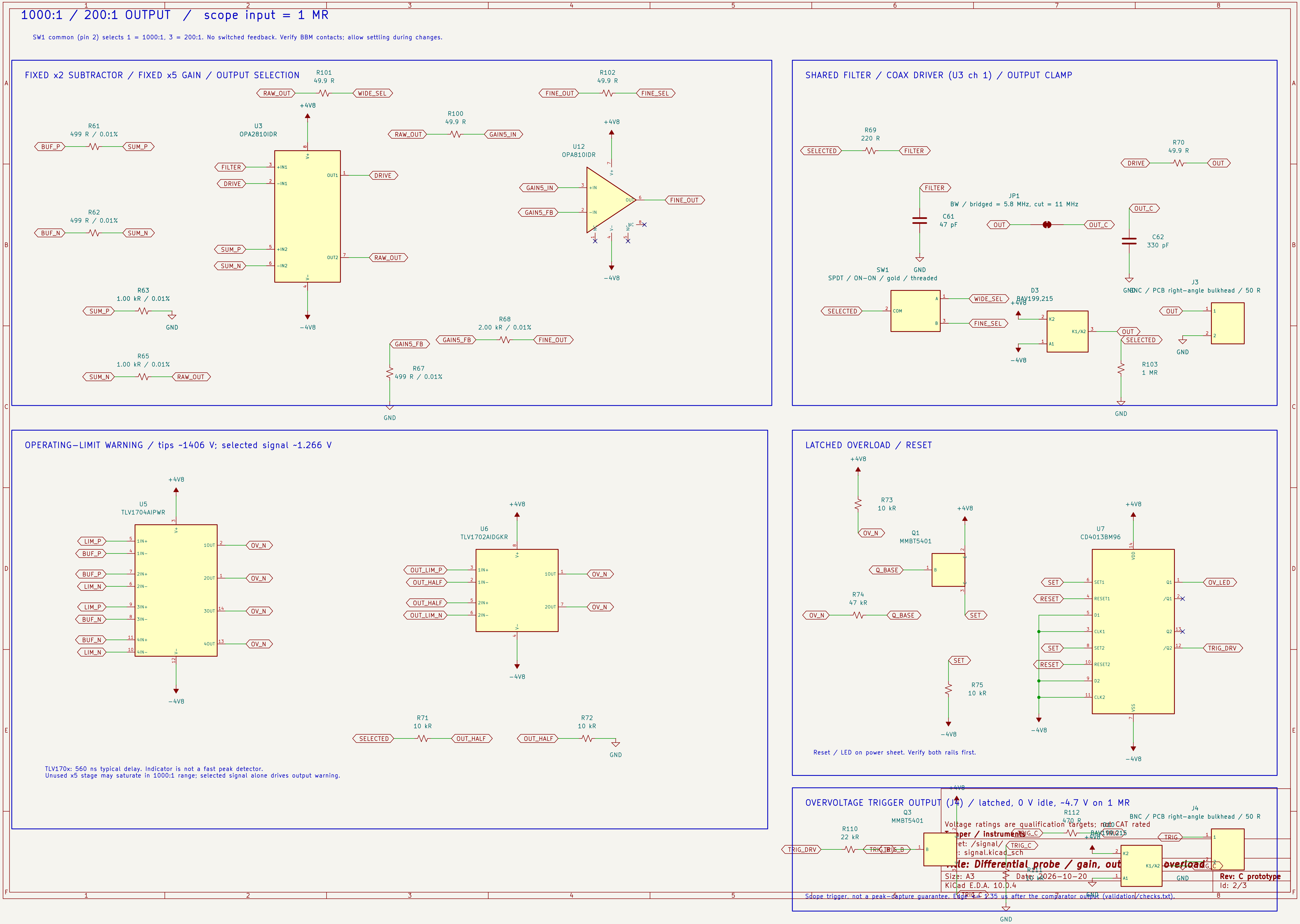
Fixed 10 nF 1% + select–on–test S1 (E24 1.6–3.9 nF, 1%)
+ S2 (E24 0–360 pF, 1%) + 12–60 pF trimmer.
Nominal 10 nF + 3.0 nF + 160 pF + ~33 pF.
CT1/2: rotor to GND. Adjust through shield tool holes.
TIP_P / TIP_N: captive HV leads.

R1
2 MR / 1% / 3 kV
R2
2 MR / 1% / 3 kV
R3
2 MR / 1% / 3 kV
R4
2 MR / 1% / 3 kV
R5
2 MR / 1% / 3 kV
TIP_P
P_SEG1
P_SEG2
P_SEG3
P_SEG4
TAP_P
C1
33 pF / 3 kV
C2
33 pF / 3 kV
C3
33 pF / 3 kV
C4
33 pF / 3 kV
C5
33 pF / 3 kV
R12
220 R
TAP_P
IN_P
R11
4.87 kR / 0.1%
TRIM_P
RV1
500 R / set 132.5 R
CCW
WIPER
CW
GND
D1
BAV199,215
+4V8
−4V8
K2
A1
K1/A2
IN_P
+4V8
U1
OPA2810IDR
IN_P
+IN1
BUF_P
−IN1
OUT1
BUF_P
IN_N
+IN2
BUF_N
−IN2
OUT2
BUF_N
V+
V−
−4V8
R21
2 MR / 1% / 3 kV
R22
2 MR / 1% / 3 kV
R23
2 MR / 1% / 3 kV
R24
2 MR / 1% / 3 kV
R25
2 MR / 1% / 3 kV
TIP_N
N_SEG1
N_SEG2
N_SEG3
N_SEG4
TAP_N
C21
33 pF / 3 kV
C22
33 pF / 3 kV
C23
33 pF / 3 kV
C24
33 pF / 3 kV
C25
33 pF / 3 kV
R32
220 R
TAP_N
IN_N
R31
4.87 kR / 0.1%
TRIM_N
RV2
500 R / set 132.5 R
CCW
WIPER
CW
GND
D2
BAV199,215
+4V8
−4V8
K2
A1
K1/A2
IN_N
TAP_P
C41
10 nF / 1%
C42
SOT 3.0 nF / 1%
C43
SOT 160 pF / 1%
GND
TAP_N
C51
10 nF / 1%
C52
SOT 3.0 nF / 1%
C53
SOT 160 pF / 1%
GND
CT1
12–60 pF / set ~33 pF
TAP_P
GND
signal
File: signal.kicad_sch
power
File: power.kicad_sch
CT2
12–60 pF / set ~33 pF
TAP_N
GND

Voltage ratings are qualification targets; not CAT rated
Temper / instruments
Sheet: /
File: differential–probe.kicad_sch

Title: Differential probe / compensated inputs

Size: A3	Date: 2026–10–20	**Rev: C prototype**
KiCad E.D.A. 10.0.4		Id: 1/3



1000:1 / 200:1 OUTPUT / scope input = 1 MR

SW1 common (pin 2) selects 1 = 1000:1, 3 = 200:1. No switched feedback. Verify BBM contacts; allow settling during changes.

FIXED x2 SUBTRACTOR / FIXED x5 GAIN / OUTPUT SELECTION

SHARED FILTER / COAX DRIVER (U3 ch 1) / OUTPUT CLAMP

OPERATING–LIMIT WARNING / tips ~1406 V; selected signal ~1.266 V

TLV170x: 560 ns typical delay. Indicator is not a fast peak detector.
Unused x5 stage may saturate in 1000:1 range; selected signal alone drives output warning.

LATCHED OVERLOAD / RESET

Reset / LED on power sheet. Verify both rails first.

OVERVOLTAGE TRIGGER OUTPUT (J4) / latched, 0 V idle, ~4.7 V on 1 MR

Voltage ratings are qualification targets; not CAT rated

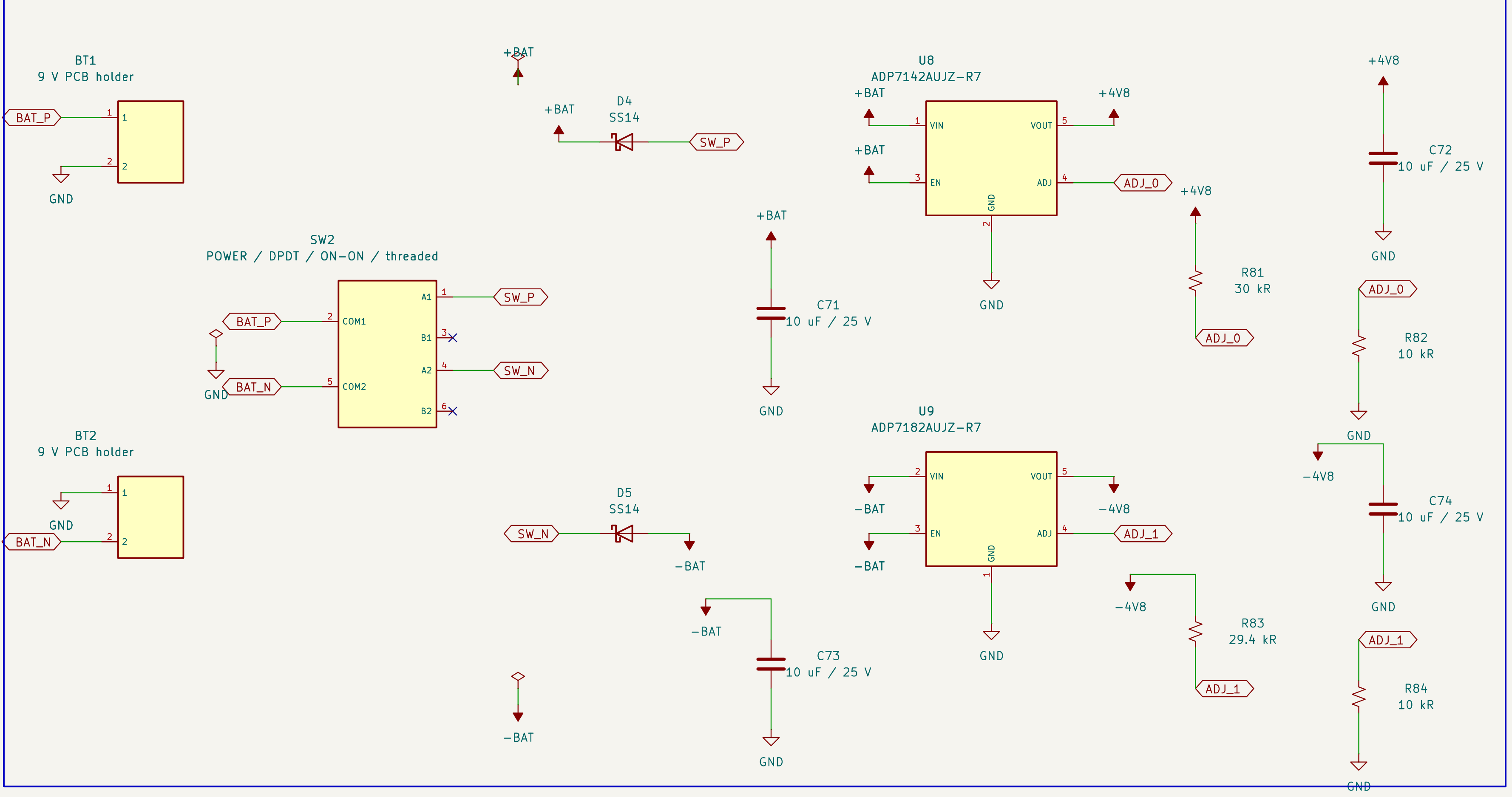
Scope trigger, not a peak–capture guarantee. Edge <= 1.35 us after the comparator output (validation/checks.txt).

Tamper / instruments
Sheet: /signal/
File: signal.kicad_sch
Title: Differential probe / gain, output, overload
Size: A3
Date: 2026–10–20
KiCad E.D.A. 10.0.4
Rev: C prototype
Id: 2/3

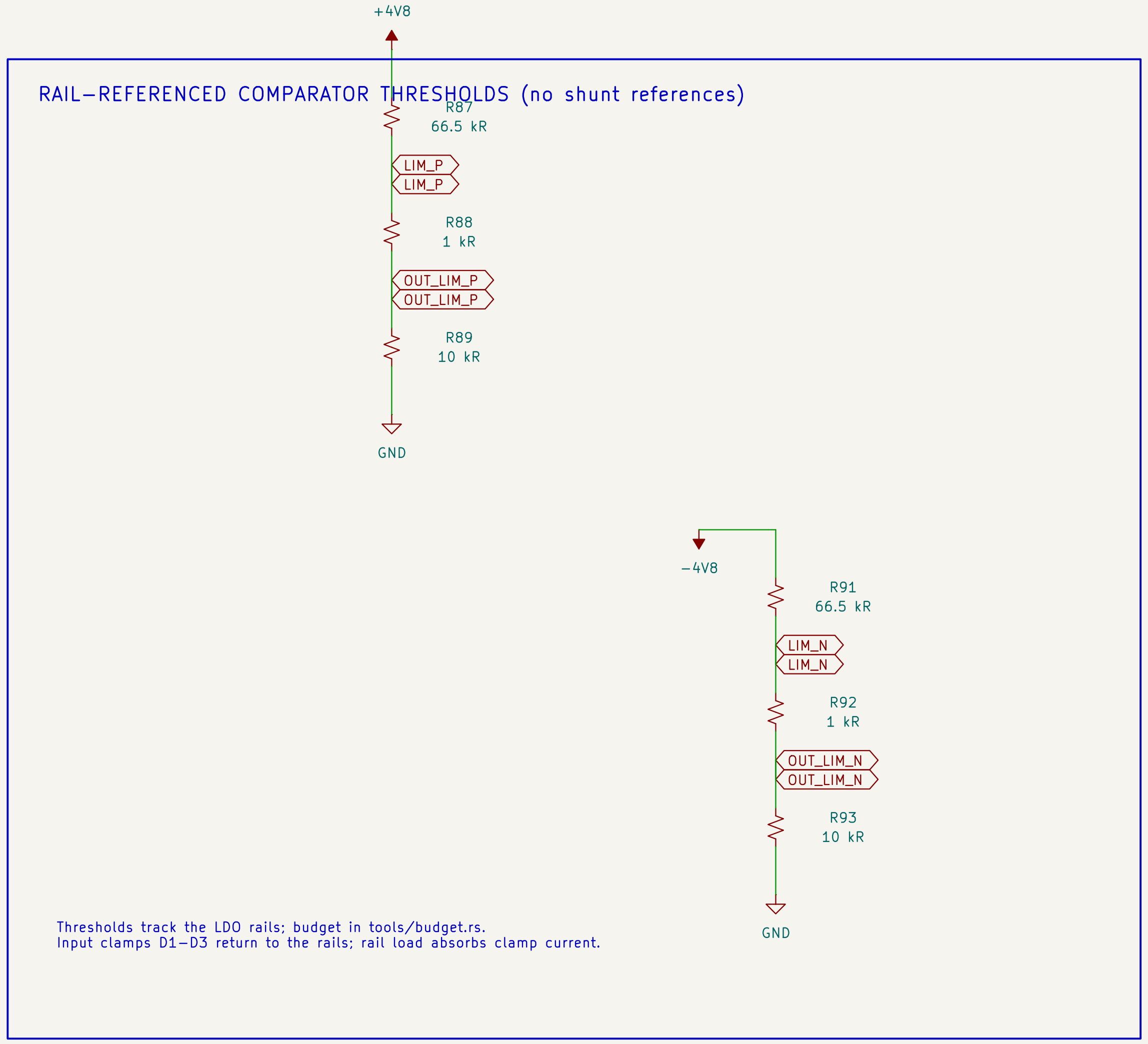
R101
49.9 R
RAW_OUT
WIDE_SEL
R102
49.9 R
FINE_OUT
FINE_SEL
R100
49.9 R
RAW_OUT
GAIN5_IN
R61
499 R / 0.01%
BUF_P
SUM_P
R62
499 R / 0.01%
BUF_N
SUM_N
U3
OPA2810IDR
+4V8
FILTER
DRIVE
+IN1
−IN1
OUT1
+IN2
−IN2
OUT2
SUM_P
SUM_N
RAW_OUT
−4V8
U12
OPA810IDR
GAIN5_IN
GAIN5_FB
+IN
−IN
OUT
V+
V−
NC
FINE_OUT
R63
1.00 kR / 0.01%
SUM_P
GND
R65
1.00 kR / 0.01%
SUM_N
RAW_OUT
R68
2.00 kR / 0.01%
GAIN5_FB
FINE_OUT
R67
499 R / 0.01%
GND
R69
220 R
SELECTED
FILTER
R70
49.9 R
DRIVE
OUT
C61
47 pF
JP1
BW / bridged = 5.8 MHz, cut = 11 MHz
OUT
OUT_C
C62
330 pF
SW1
SPDT / ON–ON / gold / threaded
COM
A
B
WIDE_SEL
FINE_SEL
D3
BAV199,215
K2
A1
K1/A2
J3
BNC / PCB right–angle bulkhead / 50 R
R103
1 MR
GND
U5
TLV1704AIPWR
LIM_P
BUF_P
BUF_N
LIM_N
1IN+
1IN−
2IN+
2IN−
3IN+
3IN−
4IN+
4IN−
1OUT
2OUT
3OUT
4OUT
OV_N
U6
TLV1702AIDGKR
OUT_LIM_P
OUT_HALF
OUT_LIM_N
R71
10 kR
SELECTED
OUT_HALF
R72
10 kR
R73
10 kR
OV_N
Q1
MMBT5401
Q_BASE
R74
47 kR
SET
R75
10 kR
U7
CD4013BM96
SET1
RESET1
D1
CLK1
SET2
RESET2
D2
CLK2
VDD
VSS
Q1
/Q1
Q2
/Q2
SET
RESET
OV_LED
TRIG_DRV
Q3
MMBT5401
R110
22 kR
TRIG_B
R111
10 kR
R112
470 R
TRIG_C
D10
BAV199,215
J4
BNC / PCB right–angle bulkhead / 50 R
TRIG

BATTERY POWER / scope shield defines circuit ground

TWO 9 V BATTERIES / +/−4.8 V LOW−NOISE RAILS

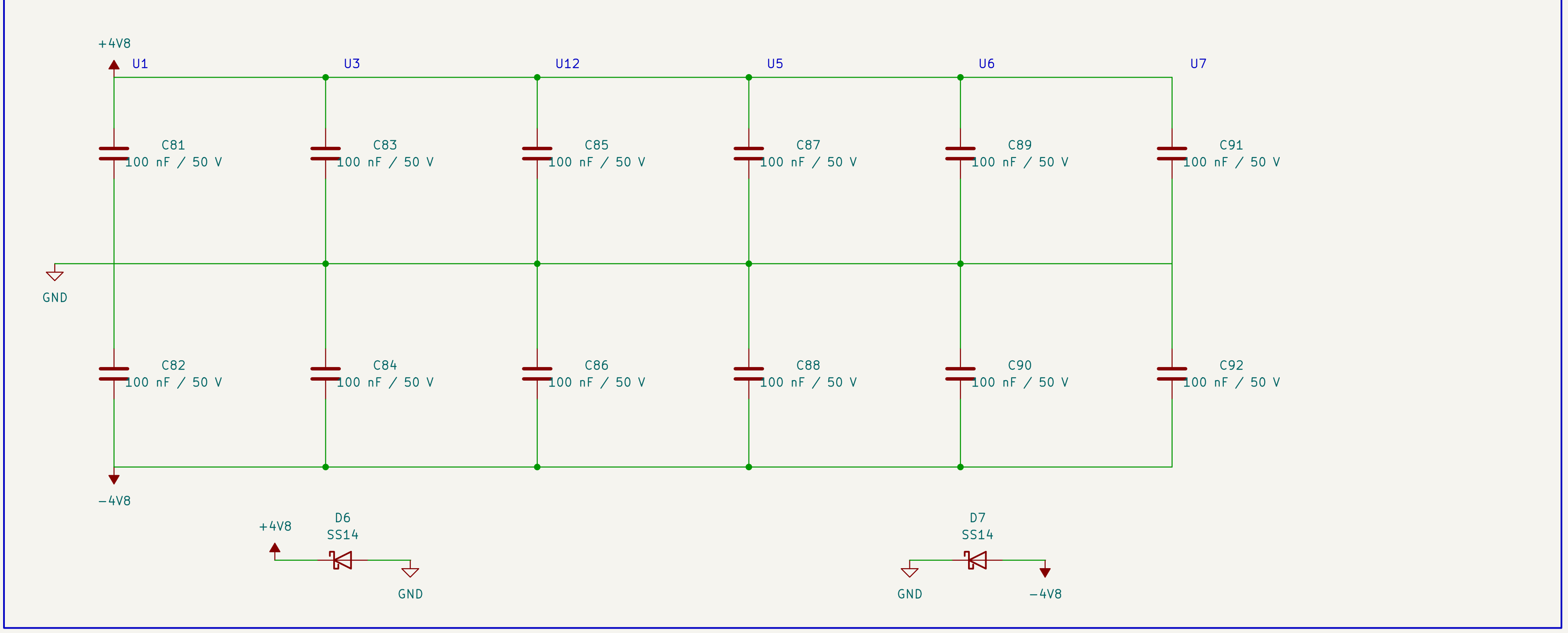

BT1
9 V PCB holder
BAT_P
GND
BT2
9 V PCB holder
BAT_N
SW2
POWER / DPDT / ON−ON / threaded
COM1
COM2
A1
B1
A2
B2
SW_P
SW_N
+BAT
D4
SS14
D5
SS14
−BAT
C71
10 uF / 25 V
C73
10 uF / 25 V
U8
ADP7142AUJZ−R7
VIN
EN
VOUT
ADJ
+4V8
ADJ_0
U9
ADP7182AUJZ−R7
−4V8
ADJ_1
R81
30 kR
R82
10 kR
R83
29.4 kR
R84
10 kR
C72
10 uF / 25 V
C74
10 uF / 25 V

RAIL−REFERENCED COMPARATOR THRESHOLDS (no shunt references)

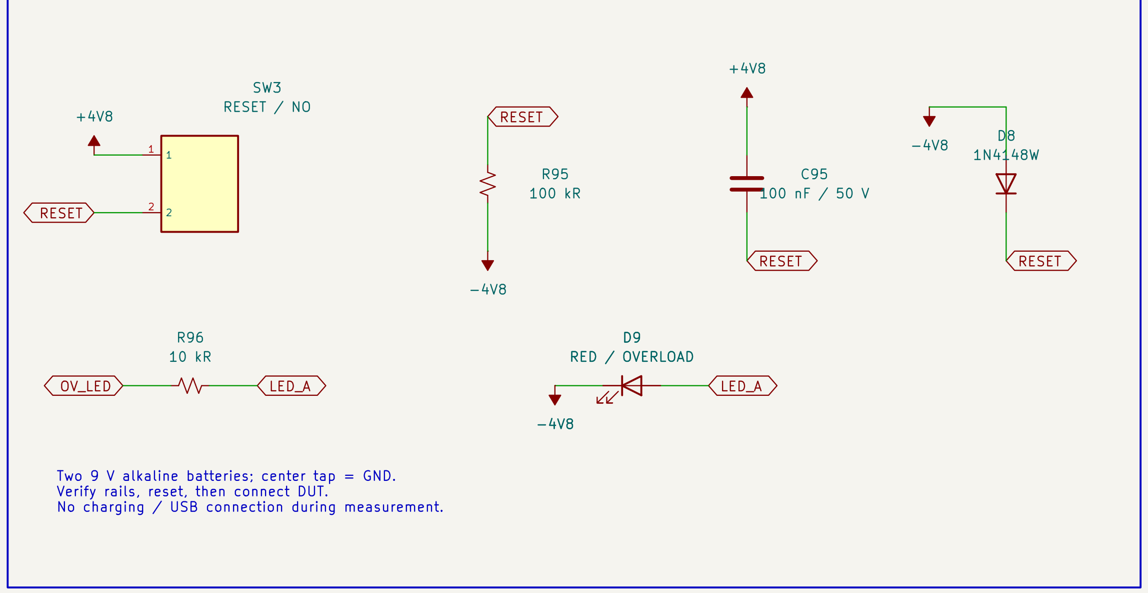

+4V8
R87
66.5 kR
LIM_P
R88
1 kR
OUT_LIM_P
R89
10 kR
GND
−4V8
R91
66.5 kR
LIM_N
R92
1 kR
OUT_LIM_N
R93
10 kR

Thresholds track the LDO rails; budget in tools/budget.rs.
Input clamps D1−D3 return to the rails; rail load absorbs clamp current.

LOCAL DECOUPLING / RAIL REVERSAL DIODES


+4V8
U1
U3
U12
U5
U6
U7
C81
C83
C85
C87
C89
C91
C82
C84
C86
C88
C90
C92
100 nF / 50 V
GND
−4V8
D6
SS14
D7
SS14

RESET / STATUS / POWER VALIDATION


SW3
RESET / NO
+4V8
RESET
R95
100 kR
−4V8
C95
100 nF / 50 V
D8
1N4148W
R96
10 kR
OV_LED
LED_A
D9
RED / OVERLOAD

Two 9 V alkaline batteries; center tap = GND.
Verify rails, reset, then connect DUT.
No charging / USB connection during measurement.

Voltage ratings are qualification targets; not CAT rated
Temper / instruments
Sheet: /power/
File: power.kicad_sch
Title: Differential probe / battery supply and thresholds
Size: A3
Date: 2026−10−20
Rev: C prototype
KiCad E.D.A. 10.0.4
Id: 3/3